GitHub Actions Setup — Workload Identity Federation

Time required: ~10 minutes Where to run: Google Cloud Shell (browser — no local install needed)

What this does

Sets up secure, key-less authentication between GitHub Actions and GCP. Once done, every push to `main` in `physioguidance1-arch/physioguidance` will automatically deploy to production without any credentials stored in GitHub.

What Ghulam needs from you

After you finish, send Ghulam these two values:

- **WIF Provider path** (a long string starting with `projects/59901490054/...`)
- **Service Account email** (`github-actions-deployer@nelc-elearning-platform-479319.iam.gserviceaccount.com`)

He'll use them to configure the GitHub Actions workflows.

Step-by-step

1. Open Cloud Shell

Go to: https://console.cloud.google.com/?project=nelc-elearning-platform-479319

Click the `>_` **Cloud Shell icon** in the top-right corner. A terminal opens at the bottom of the browser.

Confirm the project is set:

```
gcloud config get-value project
```

Expected output: `nelc-elearning-platform-479319`

If not, set it:

```
gcloud config set project nelc-elearning-platform-479319
```

2. Create the Workload Identity Pool

```
gcloud iam workload-identity-pools create github-actions-pool \
  --project=nelc-elearning-platform-479319 \
  --location=global \
  --display-name="GitHub Actions Pool"
```

Expected: `Created workload identity pool [github-actions-pool].`

3. Create the OIDC Provider (linked to GitHub)

```
gcloud iam workload-identity-pools providers create-oidc github-provider \
  --project=nelc-elearning-platform-479319 \
  --location=global \
  --workload-identity-pool=github-actions-pool \
  --display-name="GitHub OIDC" \
  --attribute-mapping="google.subject=assertion.sub,attribute.repository=assertion.repository,attribute.repository_owner=assertion.repository_owner" \
  --attribute-condition="assertion.repository_owner == 'physioguidance1-arch'" \
  --issuer-uri="https://token.actions.githubusercontent.com"
```

Expected: `Created workload identity pool provider [github-provider].`

The `--attribute-condition` line is critical — it locks WIF so only workflows running in the `physioguidance1-arch` org can authenticate.

4. Create the deploy service account

```
gcloud iam service-accounts create github-actions-deployer \
  --project=nelc-elearning-platform-479319 \
  --display-name="GitHub Actions Deployer"
```

Expected: `Created service account [github-actions-deployer].`

5. Grant the deploy service account 4 IAM roles

Run all 4 in one block:

```
SA="github-actions-deployer@nelc-elearning-platform-479319.iam.gserviceaccount.com"
PROJECT="nelc-elearning-platform-479319"

for ROLE in roles/run.admin roles/artifactregistry.writer roles/iam.serviceAccountUser roles/cloudsql.client; do
  gcloud projects add-iam-policy-binding $PROJECT \
    --member="serviceAccount:$SA" \
    --role="$ROLE" \
    --condition=None
done
```

Expected: 4 blocks of IAM policy output.

Why these 4 roles (least-privilege):

- `run.admin` — deploy to Cloud Run
- `artifactregistry.writer` — push Docker images
- `iam.serviceAccountUser` — allow Cloud Run to run the service as its runtime SA
- `cloudsql.client` — attach Cloud SQL instances (--add-cloudsql-instances flag)

6. Bind the WIF pool to the deploy SA (only for the specific repo)

```
gcloud iam service-accounts add-iam-policy-binding \
  github-actions-deployer@nelc-elearning-platform-479319.iam.gserviceaccount.com \
  --project=nelc-elearning-platform-479319 \
  --role="roles/iam.workloadIdentityUser" \
  --member="principalSet://iam.googleapis.com/projects/59901490054/locations/global/workloadIdentityPools/github-actions-
      pool/attribute.repository/physioguidance1-arch/physioguidance"
```

Expected: `Updated IAM policy for serviceAccount ...`

This is the security lock: only the specific repo `physioguidance1-arch/physioguidance` can impersonate this SA.

7. Get the two values Ghulam needs

```
echo ""
echo "========================================="
echo "SEND THESE TWO VALUES TO GHULAM:"
echo "========================================="
echo ""
echo "WIF_PROVIDER:"
echo "projects/59901490054/locations/global/workloadIdentityPools/github-actions-pool/providers/github-provider"
echo ""
echo "SA_EMAIL:"
echo "github-actions-deployer@nelc-elearning-platform-479319.iam.gserviceaccount.com"
echo ""
echo "========================================="
```

Copy both values and send them to Ghulam.

Done!

Once Ghulam has those two values, he'll add them as GitHub Actions secrets in the `physioguidance1-arch/physioguidance` repo and wire up the deploy workflows.

Cost impact

$0. Workload Identity Federation and the service account are both free. GitHub Actions cost stays at $0 (free tier is 2000 min/mo; we'll use ~200 min/mo).

Security notes

- No long-lived credentials stored anywhere. Every deploy gets a short-lived OIDC token
- Only the specific `physioguidance1-arch/physioguidance` repo can authenticate — other repos in the org (if any) cannot
- The deploy SA has only the 4 minimum roles needed — no owner/editor access
- You can audit every deploy via Cloud Audit Logs (search for `github-actions-deployer`)

To undo (if ever needed)

```
# Delete the WIF pool (irreversible after 30 days)
gcloud iam workload-identity-pools delete github-actions-pool --location=global

# Delete the service account
gcloud iam service-accounts delete github-actions-deployer@nelc-elearning-platform-479319.iam.gserviceaccount.com
```

After deletion, GitHub Actions can no longer authenticate to GCP.